Review of revised article

Major comments

I still have several critical comments about the manuscript that I do not think were fully addressed by the responses, and I think other issues may have been introduced.

1. The estimands are not clear in the article. The authors have identified ρ as a nuisance parameter and ϕ as the causal design level quantity, which is referred to as the indirect effect. However, in the autoregressive model I do not believe that ϕ represents the indirect effect. For example, if z_j is cluster j's allocation and $z_{j'}$ the allocation of the other clusters with potential outcome $Y_i(z_j, z_{j'})$ then we might specify an indirect effect as something like $E[Y(0,\mathbf{1})] - E[Y(0,\mathbf{0})]$. If the autoregressive model is the correct model, then this quantity would include a contribution of ρ through the spatial autoregression as well as ϕ. Indeed, with $\rho \neq 0$ "control-only" spillover still includes a spillover mechanism from treated to treated clusters. It would help to be explicit about what the treatment effects are what the parameters mean, and if the model identifies them. In other places β is just referred to as "the treatment effect". In Section 2.3.1 it notes "… $\hat{\beta}$ is more precisely interpreted as a total treatment effect." This doesn't seem to be correct in the context of the model where β would be only the direct effect. Indeed (6) provides the "equilibrium" state of the spatially autoregressive model and it's clear that the expectation of any individual y_{ik} is a function of the whole mean vector and parameter ρ and that $E[Y(0,\mathbf{1})] - E[Y(0,\mathbf{0})]$ does not equal any single parameter in the model.
2. The spillover effect is unclear – the authors reference both distance and neighbour functions. The spillover effect is written as $\phi(d_i)$ where d_i is "distance from the nearest intervention source", by an intervention source is not defined. Is it the boundary of the cluster or another treated unit? If it's at individual level, then the spillover is heterogeneous across individuals. But then later it's assumed to be neighbour 1/0 only, but this would be at the level of the cluster only. Then in the model (6) ϕ is now a parameter and not a function of anything. If so, then for almost all allocations on the small grids the treatment indicator and spillover indicator will be exactly collinear and the effect not identifiable. It's also unclear because in the notation they vary at the individual level. Similarly, the study observation locations (e.g. Fig 4) are simulated to have random locations but if everything is at cluster level and not distance based then this location information is discarded, right?
3. There are several issues with the notation.
    a. (1) w_{ij} is not defined.
    b. (1) why is y indexed by both *i* and *k* but not x or ε? (3) x and z have *i,k* indexes
    c. (4)-(6) the cluster index is dropped altogether now, why?
    d. The matrix W – how is "neighbour" defined here? The entries appear to be at individual level but the neighbour status is at cluster level, and if the rows are row-normalised then really this is a spillover of the cluster mean from cluster to cluster, not individual level observations.
    e. ϕ is used both as a function and as a scalar
    f. (4) the indexing over $\mathcal{N}$ seems to be redundant since w_{ij} is zero if it's not in the set of neighbours.
    g. (5) contains a different expression that (4)
4. The simulation study remains a little unclear to me. What model is being estimated for the simulation study. While (6) gives the DGP, the results suggest that the models do not include the spillover indicator, but potentially do include the autogressive component. As per point

2, the control only spillover bias for block stratified sampling is effectively the value of ϕ which is what you'd expect with perfect collinearity and omitted variable bias. The estimated model should be explained.

5. In my first round comments I queried whether the lack of permutations of the treatment allocation made sense in the context of a randomised controlled trial. The authors' response was that "BSS is a design-stage tool evaluated through exhaustive enumeration, not a randomization distribution for permutation inference." But this does not address the issue for Frequentist inference which relies on interpretation around the sampling distribution under the null. The sampling distribution for BSS seems to have only 1 or 2 values and the treatment allocation would be almost deterministic. If it's only a design-stage tool, what purpose does it serve if it's not the actual allocation mechanism?
6. Interpretation. The results suggest that with the models described here the bias might be potentially catastrophic and so none of these designs should be used. BSS might be slightly better in some scenarios but its so severe that "fried egg" or "buffer zone" designs should really be the preferred approach. "It's so severe"??? - what is severe